

MUHAMMAD TALHA SHAIKH

Junior Full-Stack & AI Engineer

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PROFESSIONAL SUMMARY

High-velocity **Junior Full-Stack & AI Engineer** (~1.5 years experience) trained through **GIAIC (Governor Sindh Initiative for GenAI & Web3)**. Proven track record architecting and deploying scalable web systems, agentic AI pipelines, and multi-tenant platforms solo. Shipped **Botaura** (162+ route multi-tenant RAG SaaS with Meta WhatsApp Tech Provider status), **FlowCreator OS** (AI video directing engine via Google Gemini SDK), **ShieldTools** (zero-server client-side privacy & COD economics platform), and **Volvelo** (luxury European marketplace with Clerk RBAC). Focused on clean architecture, sub-second latency, and immediate engineering impact.

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript (ES6+), Python, Next.js 16 (App Router, Server Actions), React 19, FastAPI, Node.js, HTML5 Canvas API, jsPDF
AI & Vector Retrieval (RAG): Google Gemini 2.5/3.6 Flash SDK, Reference Image Anchoring, OpenAI SDK, pgvector cosine similarity, BM25 Hybrid Retrieval, LangChain, Zod Structured Outputs
Databases & Cloud: Neon PostgreSQL, Prisma ORM 6, Drizzle ORM, Supabase, Cloudflare R2, AWS S3, Vercel (Edge & Cron), Docker, Git & GitHub Actions
Integrations & Architecture: WhatsApp Cloud API (Meta Tech Provider), Clerk Auth (RBAC Sync), Stripe Checkout, Resend, PWA, Zero-Knowledge Client Architecture, Multi-Tenant Row-Level Security

FEATURED SYSTEMS & ENGINEERING PROJECTS

Botaura — Multi-Tenant RAG AI Chatbot & WhatsApp Commerce SaaS [Flagship] Founder & Solo Engineer

Live: <https://botaura.app> · Case Study: talhawebsite.xyz/projects/botaura

- Architected and deployed a multi-tenant SaaS across **162+ production routes**, enabling SMBs to train custom AI chatbots and automate WhatsApp sales.
- Implemented Hybrid RAG pipeline combining OpenAI embeddings with pgvector cosine similarity and BM25 search for English & Roman-Urdu queries with confidence thresholds.
- Meta WhatsApp Tech Provider: engineered token routing by phone_number_id, handling webhooks, templates, and ad attribution across merchant accounts with zero secret exposure.
- Enforced strict server-side tenant isolation with signed JWTs, UUID cascading, and hashed API keys, preventing cross-tenant data leaks by design.

FlowCreator OS — Autonomous AI Video Directing & CreatorOps Operating System Lead System Architect

Live: <https://flow-creator-os.vercel.app/> · Case Study: talhawebsite.xyz/projects/flow-creator-os

- Engineered Autonomous AI Director using Google Gemini Flash SDK to convert creative premises into camera-sequenced video prompt bundles for Google Veo 2 and Sora.
- Designed Reference Image Anchoring injecting master biometric tokens to eliminate diffusion face-drift, and Temporal Choreography for continuous room-tone match-cuts.
- Constructed CreatorOps WhatsApp bot with PKT time-zoned cron reminders via Green-API & CallMeBot for daily multi-platform upload tracking.

ShieldTools — Zero-Server Digital Privacy & COD E-Commerce Economics Platform Lead Architect & Creator

Live: <https://tools.talhawebsite.xyz/> · Case Study: talhawebsite.xyz/projects/shield-tools

- Built 100% in-browser zero-knowledge watermarking studio (DocShield™) using HTML5 Canvas & jsPDF to watermark CNIC/Passports in RAM with \$0 backend cost.
- Engineered EcomShield™ COD unit economics engine isolating reverse courier penalties, packaging waste, and wasted ad spend on COD returns with automated WhatsApp summaries.
- Shipped installable Progressive Web App (PWA) with early prompt capture, achieving sub-second load times across 22 statically prerendered routes.

Volvelo — Luxury Multi-Tenant European Dropshipping Marketplace Full-Stack Architect

Live: <https://vovelo.vercel.app/> · Case Study: talhawebsite.xyz/projects/vovelo

- Architected 48-route platform separating customer storefront, merchant atelier portal (/portal), and super-admin hub (/admin) on Neon PostgreSQL via Prisma ORM.
- Built bidirectional Clerk Cloud RBAC synchronization with Neon PostgreSQL, ensuring instant permission propagation to user session claims without re-login.
- Engineered dynamic client-side multi-currency engine (EUR, USD, GBP, PKR) with Zustand persistence and SSR hydration guards.

Education & Credentials

Cloud Native Applied Generative AI & Full-Stack Engineering — Governor Sindh Initiative (GIAIC) Ongoing
Meta WhatsApp Tech Provider Accreditation — Verified Cloud API Integration Architecture Verified
Intensive Project-Based Systems Development — Real-world production full-stack & AI software 2022 – Present